

Quiz # 03
AI (CS-3002)

Q1: For the given statements, give your answer as true or false to show your agreement or disapproval to the following statements along with one sentence justification. No points will be awarded without justification. [02 Marks]

- i) Prepruning in decision trees involves bottom-up removal of subtrees after full tree construction.
- ii) Entropy computation in information gain is based solely on the probability of the majority class.

Q.2 The following dataset provides information about websites and their attributes. The goal is to classify whether a website is phishing (P) or legitimate (L) based on attributes such as Domain Age, HTTPS, and Suspicious Links.

S.No	Domain Age	HTTPS	Suspicious Links	Website Type
1	Old	Yes	No	L
2	New	No	Yes	P
3	Old	Yes	Yes	P
4	New	No	No	L
5	Old	Yes	Yes	P
6	New	Yes	No	L

Determine the root node of the decision tree based on the Decision Tree Algorithm (CART). [03 Marks]

Q3: Please provide your answers in **2 lines or fewer**. Answers longer than 2 lines will receive **zero marks**.

a) **import seaborn as sns**

[2 Marks]

import matplotlib.pyplot as plt

from sklearn.neighbors import KNeighborsClassifier

from sklearn.model_selection import train_test_split

df = pd.read_csv('abc_logs.csv')

correlation_matrix = df.corr()

sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

plt.show()

X = df.drop('target', axis=1)

y = df['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

knn_model = KNeighborsClassifier(n_neighbors=5)

knn_model.fit(X_train, y_train)

y_pred = knn_model.predict(X_test)

print(f'Accuracy: {accuracy_score(y_test, y_pred)}')

- i) What does the `df.corr()` function do?
- ii) How does the `KNeighborsClassifier()` work in this code, and what is the purpose of `n_neighbors=5`?
- iii) What does the `train_test_split()` function do, and why is it important for evaluating the model?

Q.4 You work as a cybersecurity analyst for a company, and your task is to identify potential anomalies in network traffic. The company monitors the network traffic based on the following metrics: [3 Marks]

1. Bytes sent (x): The number of bytes sent from the device.
2. Connections made (y): The number of connections made from the device to external sources.

Based on past traffic data, you have the following observations of 10 devices within the network:

Device ID	Bytes sent (x)	Connections made (y)
1	100	10
2	110	12
3	120	15
4	150	18
5	160	20
6	180	25
7	200	30
8	220	35
9	250	40
10	300	50

You are using K-means clustering with 3 clusters to detect any potential anomalies in the network traffic. The process involves the following steps:

1. Initial Centroids: Assume that you randomly initialize the centroids.
2. Step 1: Calculate the Euclidean distance between each data point and the three centroids.
3. Step 2: Assign each data point to the nearest centroid (cluster).
4. Step 3: Update the centroids by finding the average of the points in each cluster.
5. Step 2: Repeat steps 1 to 3 just 2 times.